Lab 1 - Greedy heuristics

Description of the problem

Each instance contains nodes with coordinates (x, y) and a cost. Select exactly half of the nodes (rounded up) and connect them into a closed cycle. Minimise **cycle length + sum of the costs of the selected nodes**. Distances are Euclidean, rounded to the nearest integer.

Algorithms

200 solutions per greedy method (one starting from each node) and 200 random solutions. "Nearest" always means the best increase of the objective (distance change + node cost), not distance alone.

```
Random:
  tour = first k nodes of a random permutation

Nearest neighbour, end only (start):
  tour = [start]
  while |tour| < k:
      u = unvisited node minimising dist(last, u) + cost(u)
      append u

Nearest neighbour, any position (start):
  tour = [start]
  while |tour| < k:
      for each unvisited u and each position p (front, between two nodes, end):
          delta = added path length + cost(u)
      insert the (u, p) with the smallest delta

Greedy cycle (start):
  v = unvisited node minimising dist(start, v) + cost(v);  tour = [start, v]
  while |tour| < k:
      for each unvisited u and each cycle edge (a, b):
          delta = d(a, u) + d(u, b) − d(a, b) + cost(u)
      insert the (u, edge) with the smallest delta
```

Results

Objective = cycle length + sum of node costs, over 200 runs per method.
All best solutions were checked with the solution checker (the same computation as the lecturer's Solution checker.xlsx): **yes**.
Results are identical in all implemented languages: **yes**.

Instance TSPA

Method	Min	Max	Avg	Best solution checked
Random solution	237131	287973	263514	yes
Nearest neighbor (end only)	84362	90075	87477	yes
Nearest neighbor (any position)	72328	77022	74347	yes
Greedy cycle	72590	75281	73797	yes

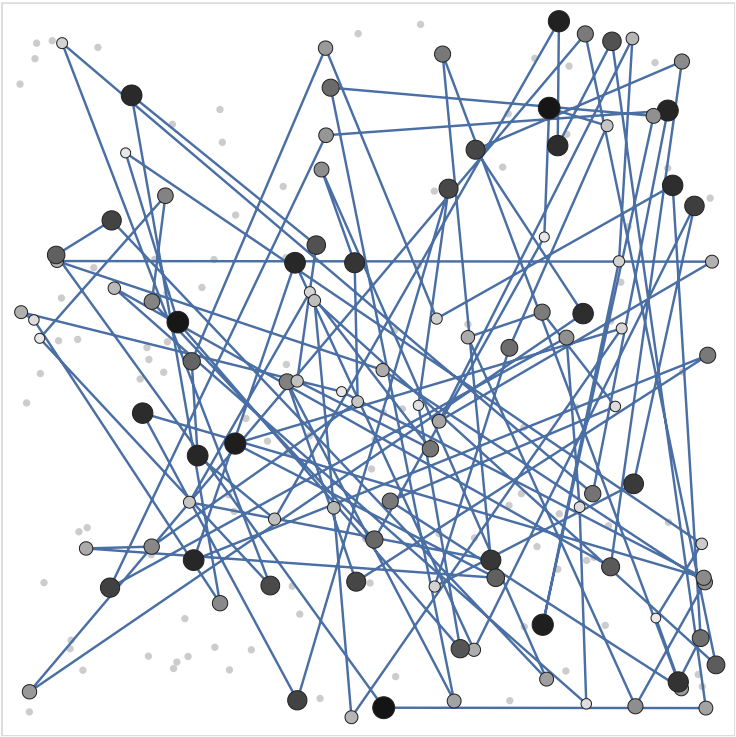
Instance TSPB

Method	Min	Max	Avg	Best solution checked
Random solution	188214	231879	206802	yes
Nearest neighbor (end only)	56007	63166	59397	yes
Nearest neighbor (any position)	52915	66164	59475	yes
Greedy cycle	53016	63736	59709	yes

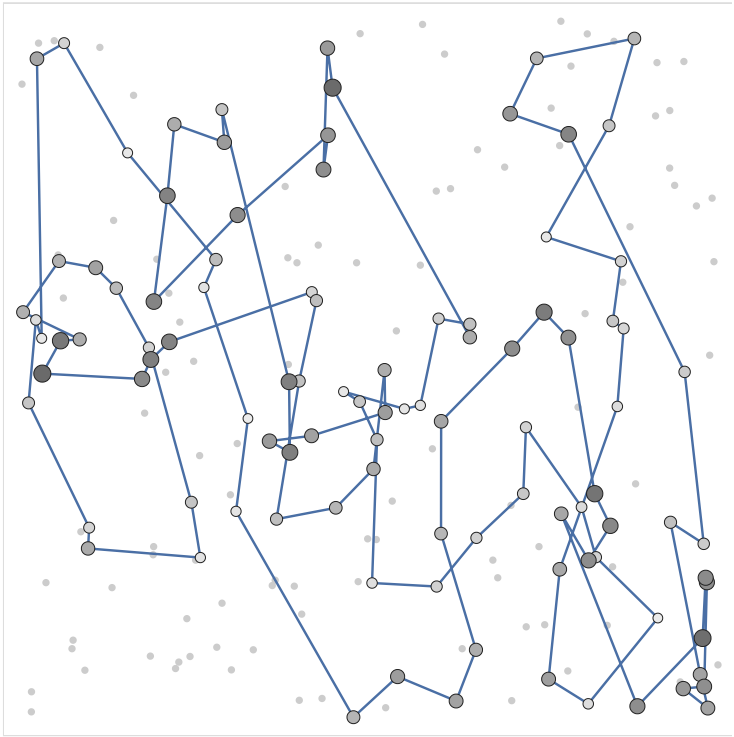
Visualisation of the best solutions

Node colour and size show the node cost (dark and large = expensive). Grey dots are nodes not selected.

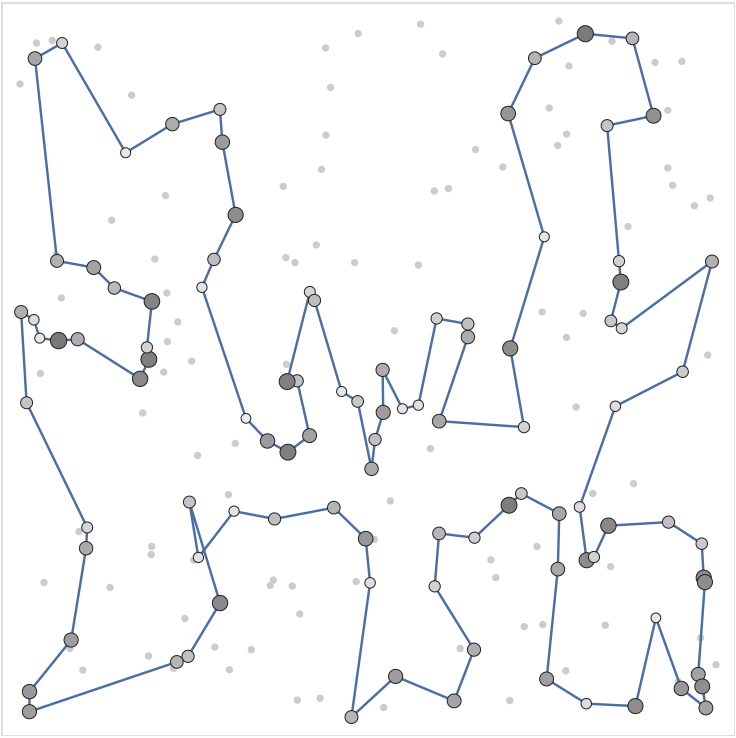
Instance TSPA



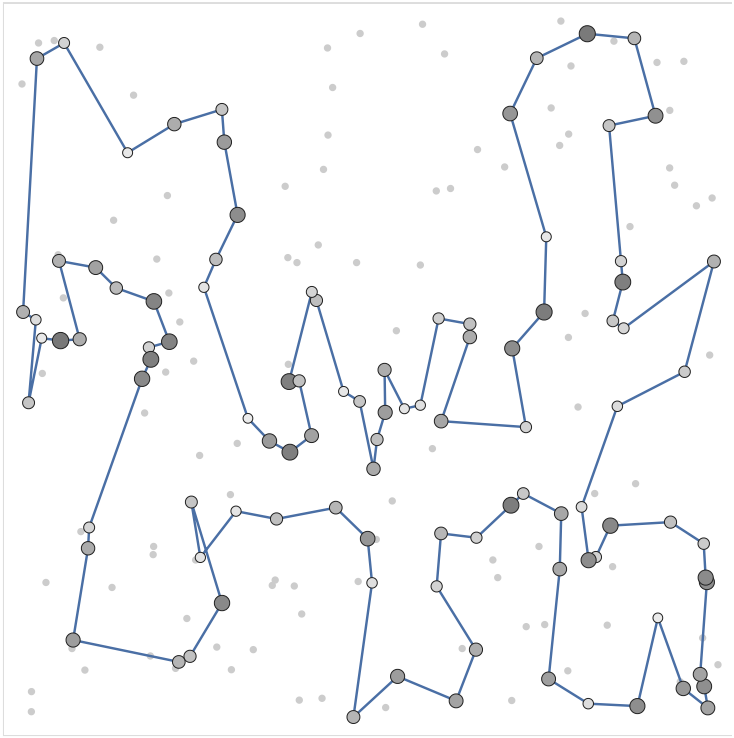
Random solution: objective 237131 (length 140636 + cost 96495)



Nearest neighbor (end only): objective 84362 (length 31319 + cost 53043)

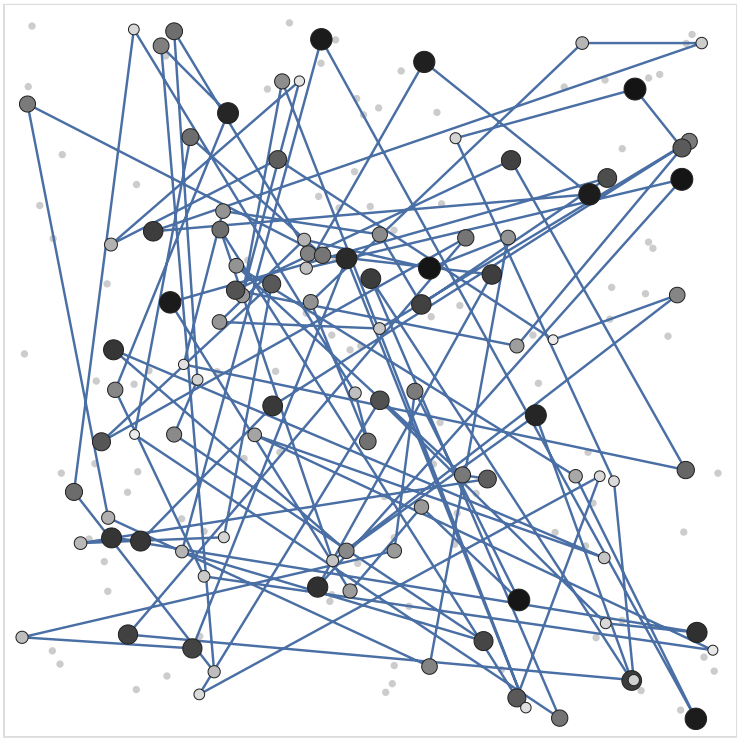


Nearest neighbor (any position): objective 72328 (length 21816 + cost 50512)

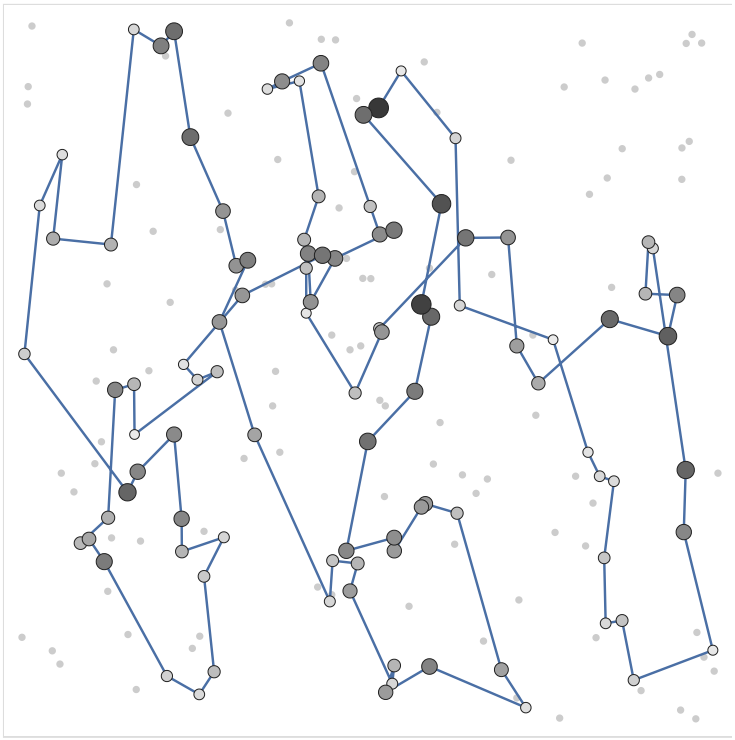


Greedy cycle: objective 72590 (length 21570 + cost 51020)

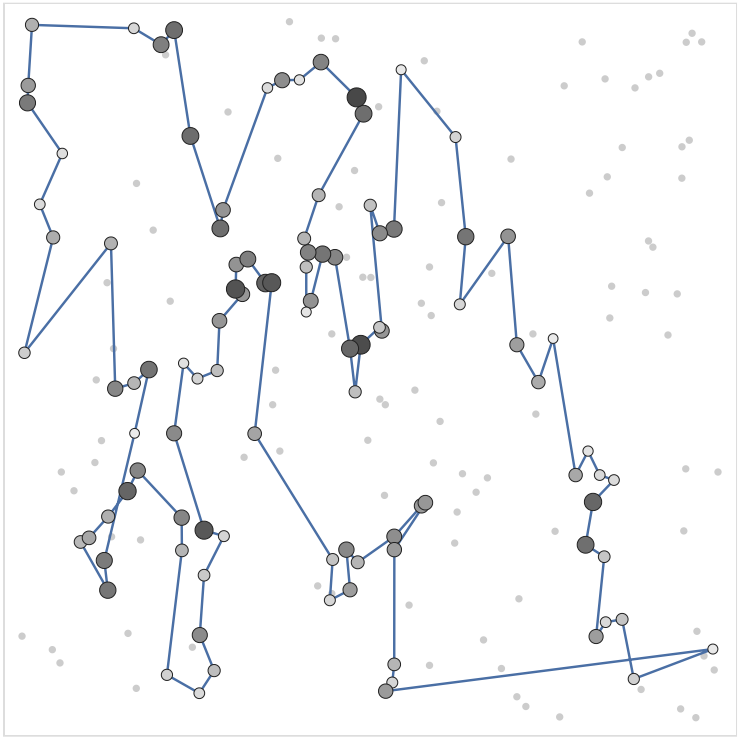
Instance TSPB



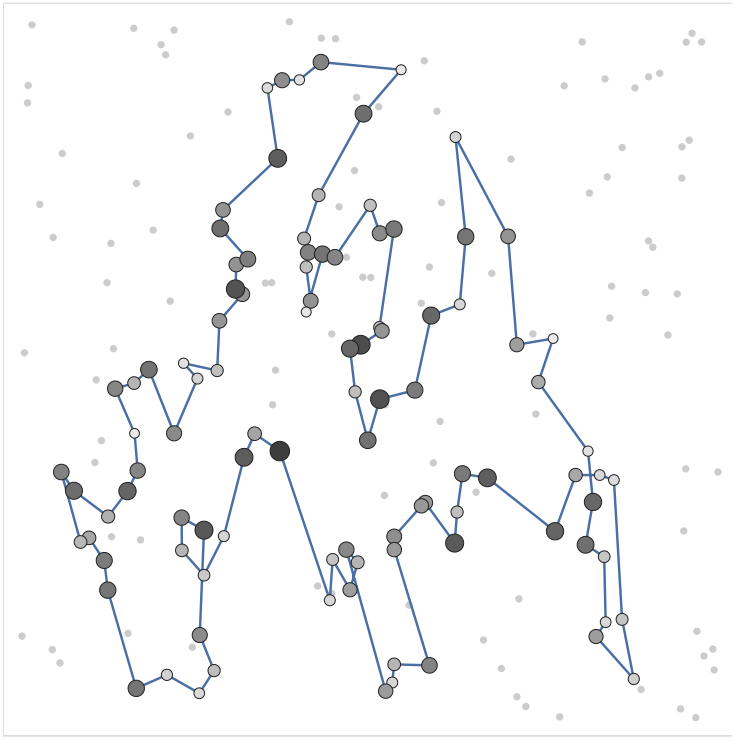
Random solution: objective 188214 (length 138112 + cost 50102)



Nearest neighbor (end only): objective 56007 (length 24632 + cost 31375)



Nearest neighbor (any position): objective 52915 (length 20068 + cost 32847)



Greedy cycle: objective 53016 (length 16704 + cost 36312)

Best solutions (0-based node indices)

Instance TSPA

Random solution (objective 237131):
128, 34, 55, 103, 152, 57, 106, 85, 120, 112, 86, 43, 48, 96, 63, 38, 100, 61, 164, 195, 178, 194, 58, 117, 6, 158, 173, 40, 37, 121, 156, 192, 5, 101, 165, 1, 29, 131, 147, 60, 52, 170, 64, 54, 142, 93, 41, 88, 107, 159, 80, 68, 20, 36, 15, 22, 12, 146, 67, 116, 149, 189, 137, 133, 35, 99, 51, 30, 13, 154, 122, 75, 155, 33, 115, 162, 23, 87, 197, 56, 19, 78, 118, 82, 66, 182, 139, 190, 144, 161, 127, 65, 191, 50, 114, 45, 16, 26, 163, 129

Nearest neighbor (end only) (objective 84362):
54, 9, 33, 36, 177, 67, 144, 53, 135, 58, 134, 176, 174, 106, 89, 107, 57, 182, 151, 93, 50, 3, 1, 164, 178, 29, 136, 186, 184, 47, 159, 130, 153, 78, 88, 193, 170, 146, 75, 64, 150, 42, 113, 43, 85, 142, 8, 131, 84, 158, 192, 181, 114, 141, 180, 132, 76, 80, 48, 31, 63, 149, 168, 97, 127, 187, 30, 123, 152, 108, 60, 116, 49, 11, 119, 160, 83, 90, 51, 16, 35, 87, 45, 140, 52, 179, 185, 28, 102, 156, 157, 71, 91, 38, 100, 4, 112, 139, 115, 5

Nearest neighbor (any position) (objective 72328):
104, 129, 69, 127, 97, 168, 116, 149, 63, 11, 49, 160, 83, 123, 100, 152, 108, 60, 31, 48, 80, 91, 157, 71, 4, 76, 132, 180, 28, 102, 185, 35, 156, 51, 16, 67, 144, 140, 53, 179, 52, 177, 36, 33, 9, 54, 131, 174, 8, 164, 186, 136, 163, 29, 101, 178, 1, 169, 3, 50, 147, 47, 93, 106, 42, 89, 113, 130, 159, 146, 170, 153, 193, 78, 88, 107, 64, 57, 182, 151, 150, 176, 72, 134, 84, 58, 158, 192, 181, 114, 135, 73, 45, 87, 141, 187, 30, 189, 2, 171

Greedy cycle (objective 72590):
31, 116, 149, 168, 63, 11, 49, 60, 108, 152, 100, 90, 123, 83, 160, 97, 127, 69, 171, 2, 189, 30, 187, 141, 87, 45, 73, 135, 114, 181, 192, 158, 58, 84, 134, 72, 176, 150, 151, 182, 57, 64, 107, 88, 78, 193, 153, 170, 146, 159, 130, 113, 89, 42, 106, 93, 47, 147, 50, 3, 169, 1, 178, 101, 29, 163, 136, 186, 164, 142, 8, 174, 131, 54, 9, 33, 36, 177, 52, 179, 53, 140, 144, 67, 16, 51, 156, 35, 185, 102, 28, 180, 132, 76, 4, 71, 157, 91, 80, 48

Instance TSPB

Random solution (objective 188214):
80, 23, 16, 71, 163, 97, 140, 42, 136, 144, 107, 193, 172, 44, 12, 15, 105, 190, 160, 98, 151, 116, 103, 45, 147, 195, 177, 60, 110, 64, 153, 55, 73, 106, 188, 137, 17, 170, 50, 8, 13, 6, 88, 104, 0, 133, 38, 184, 61, 4, 113, 5, 68, 20, 111, 134, 129, 39, 191, 59, 148, 54, 63, 158, 96, 21, 198, 171, 95, 142, 1, 181, 27, 186, 34, 51, 161, 18, 75, 46, 130, 154, 56, 77, 176, 162, 139, 199, 57, 90, 2, 91, 185, 19, 168, 40, 132, 84, 66, 30

Nearest neighbor (end only) (objective 56007):
32, 89, 57, 76, 54, 127, 129, 90, 153, 20, 14, 2, 110, 118, 56, 114, 196, 10, 148, 143, 78, 192, 162, 104, 185, 184, 3, 151, 69, 172, 133, 120, 96, 121, 198, 48, 72, 46, 86, 117, 160, 136, 8, 21, 154, 140, 47, 170, 125, 177, 6, 132, 29, 197, 175, 134, 45, 60, 40, 71, 155, 97, 115, 180, 65, 28, 149, 145, 158, 116, 147, 42, 50, 130, 30, 183, 38, 64, 85, 59, 167, 51, 87, 179, 82, 88, 171, 99, 67, 24, 34, 181, 152, 73, 98, 27, 135, 75, 79, 156

Nearest neighbor (any position) (objective 52915):
110, 2, 14, 20, 74, 153, 182, 52, 90, 129, 127, 80, 54, 192, 162, 104, 76, 185, 57, 89, 86, 46, 72, 3, 184, 173, 151, 122, 117, 8, 160, 69, 172, 136, 133, 120, 156, 43, 48, 96, 198, 121, 130, 191, 50, 42, 147, 116, 37, 146, 13, 149, 28, 145, 65, 158, 177, 125, 26, 170, 197, 36, 132, 29, 6, 180, 115, 155, 71, 175, 134, 45, 157, 60, 40, 141, 97, 154, 140, 47, 38, 21, 176, 30, 183, 189, 168, 64, 59, 85, 51, 73, 167, 152, 34, 24, 181, 179, 87, 82

Greedy cycle (objective 53016):
49, 64, 33, 40, 60, 71, 155, 141, 157, 45, 134, 175, 62, 36, 197, 29, 132, 178, 103, 6, 180, 115, 170, 177, 125, 26, 97, 140, 154, 47, 38, 21, 176, 30, 183, 191, 130, 63, 121, 198, 96, 48, 89, 156, 120, 133, 136, 172, 160, 69, 8, 117, 72, 46, 86, 184, 3, 173, 122, 151, 98, 111, 27, 135, 76, 185, 57, 104, 162, 54, 192, 127, 52, 182, 153, 20, 74, 2, 14, 90, 129, 80, 150, 84, 66, 67, 31, 24, 34, 152, 181, 88, 179, 87, 82, 73, 167, 51, 59, 85

Source code

Source code of this lab: labs/lab1 (Python, C, C++, Rust).

Conclusions

- Random solutions are about 3.5 times worse than any greedy method.
- On TSPA, inserting at any position (nearest neighbour any position, greedy cycle) is clearly better than appending at the end (average about 74000 vs 87500).
- On TSPA, nearest neighbour (any position) and greedy cycle are close (best 72328 vs 72590).
- On TSPB all three greedy methods have similar averages (about 59500).
- Greedy methods are deterministic given the start node, so the spread comes only from the start choice; trying every start and keeping the best is cheap and worthwhile.